

COMP9334: Capacity Planning of Computer Systems and Networks

Optimisation (4):
Placement problems

Integer Programming - What have you seen?

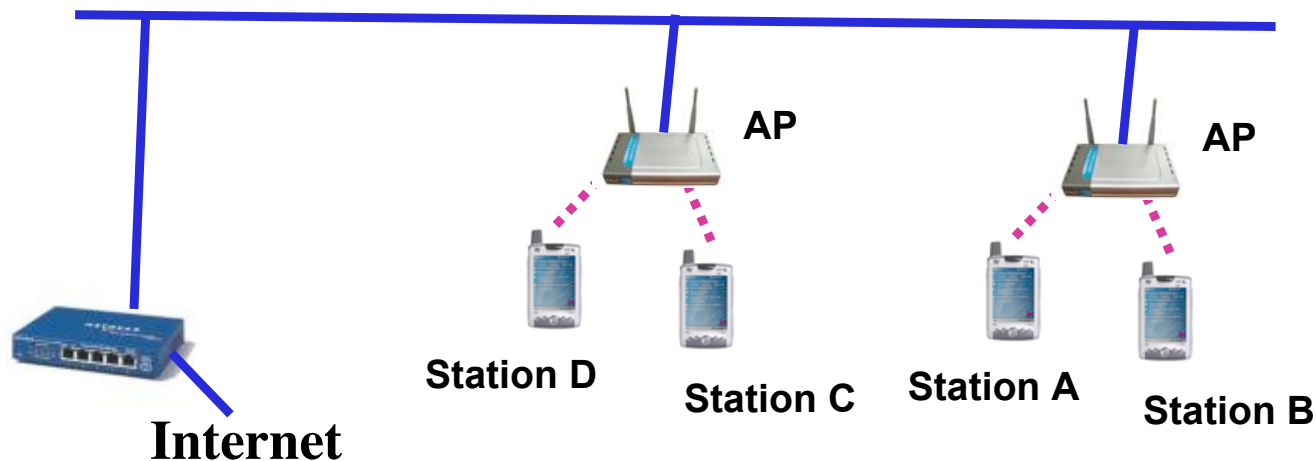
- A recurrent theme is to use integer programming to make *yes or no decisions*
- Examples of binary decisions
 - Week 9A: Grid computing problem
 - Choose a particular grid computing company or not
 - Week 9B: Routing of flows
 - Should the flow be routed on a link or not?

This week's lecture

- Integer programming for placement problem
 - Example: There are a number of potential places that I can put certain devices, what are the best places to put them?
- We will study a placement problem in wireless networks
 - Placement of wireless access points
- For the revision problem, we will look at the controller placement problem for software-defined networking

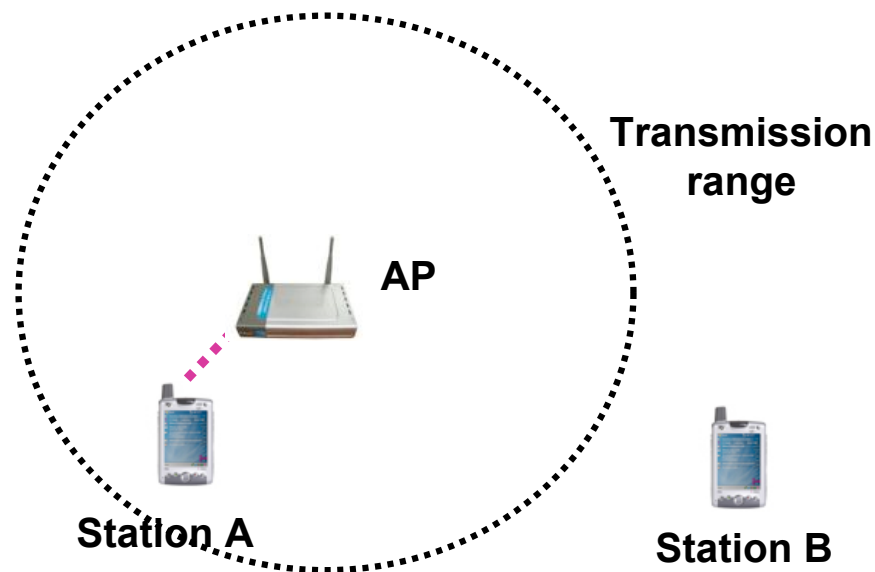
Wireless Local Area Networks

- Commonly known as Wireless LAN, WiFi etc.
- Formal standards in IEEE 802.11, IEEE 802.11a/b/g/n/ac/ad etc.
- Infrastructure mode: Wireless Access Points (APs) and Wireless stations



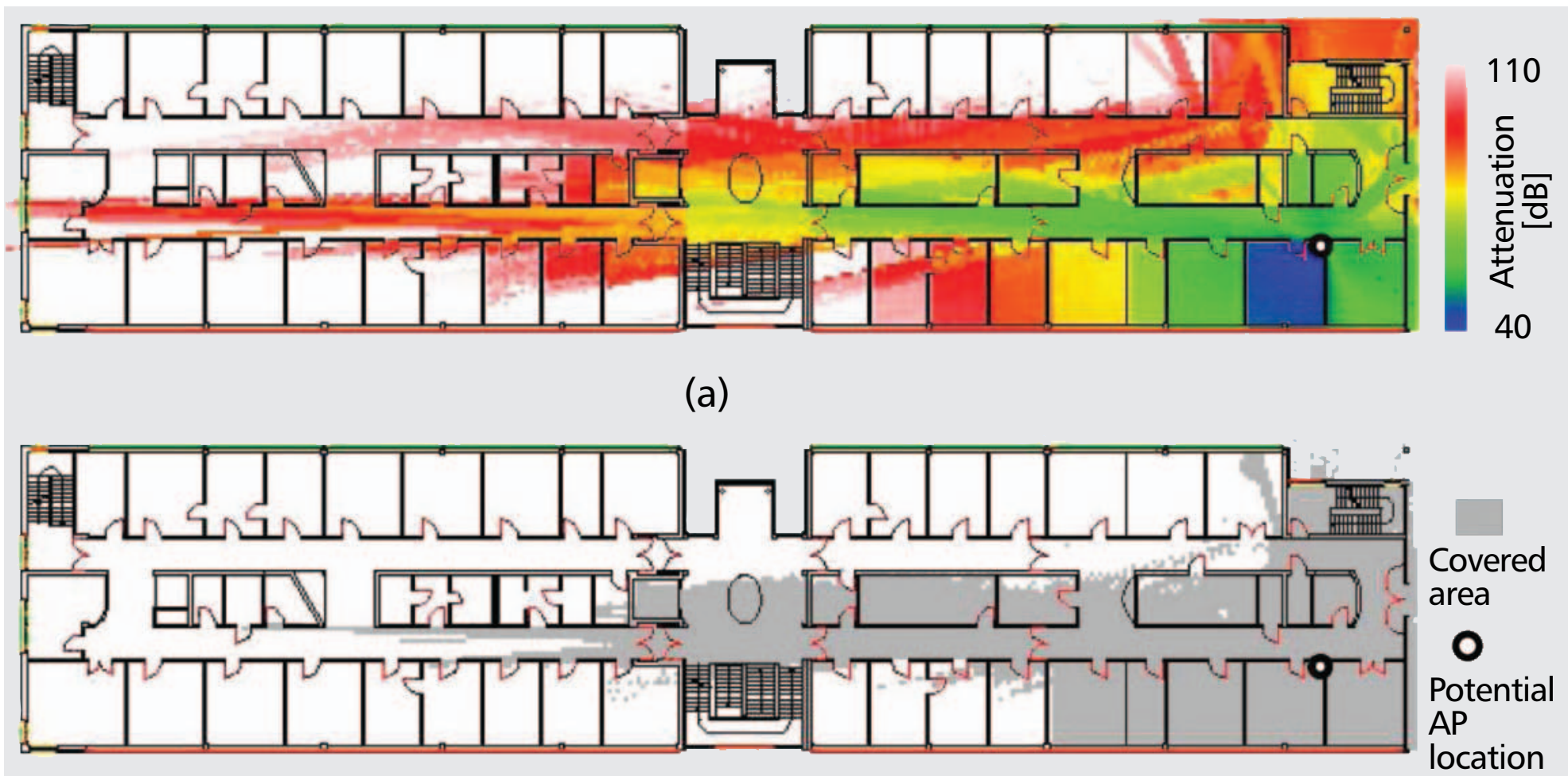
Wireless Access Point Coverage

- Due to radio propagation loss and mandated limit on transmission power, wireless access points have only limited coverage
- For example, a Cisco Aironet access point has a coverage of 304m when operating outdoor at 11 Mbps
- Ideal coverage area is a circle. In the picture below, Station A can talk to the access point but not Station B



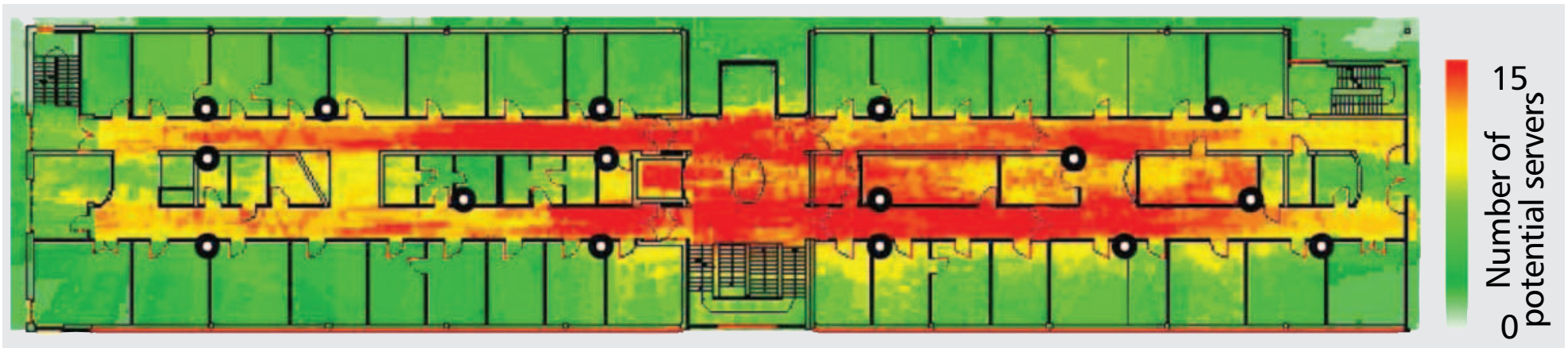
Coverage in practice

- Note: High attenuation = Poor signal = Poor coverage
- An area is covered only if the attenuation is smaller than a threshold



Covering a given area

- Multiple access points may be required to cover a given area
- Decisions to make
 - The number of access points required to cover the given area
 - The position of the access points



The coverage problem - definition

■ Given

- A number of potential access point locations $L_1, L_2, \dots, L_p$
- A number of stations $s_1, s_2, s_3, \dots, s_n$
- Binary constant δ_{ij} where

$$\delta_{ij} = \begin{cases} 1 & \text{if station } s_i \text{ is covered by an AP at location } L_j \\ 0 & \text{otherwise} \end{cases}$$

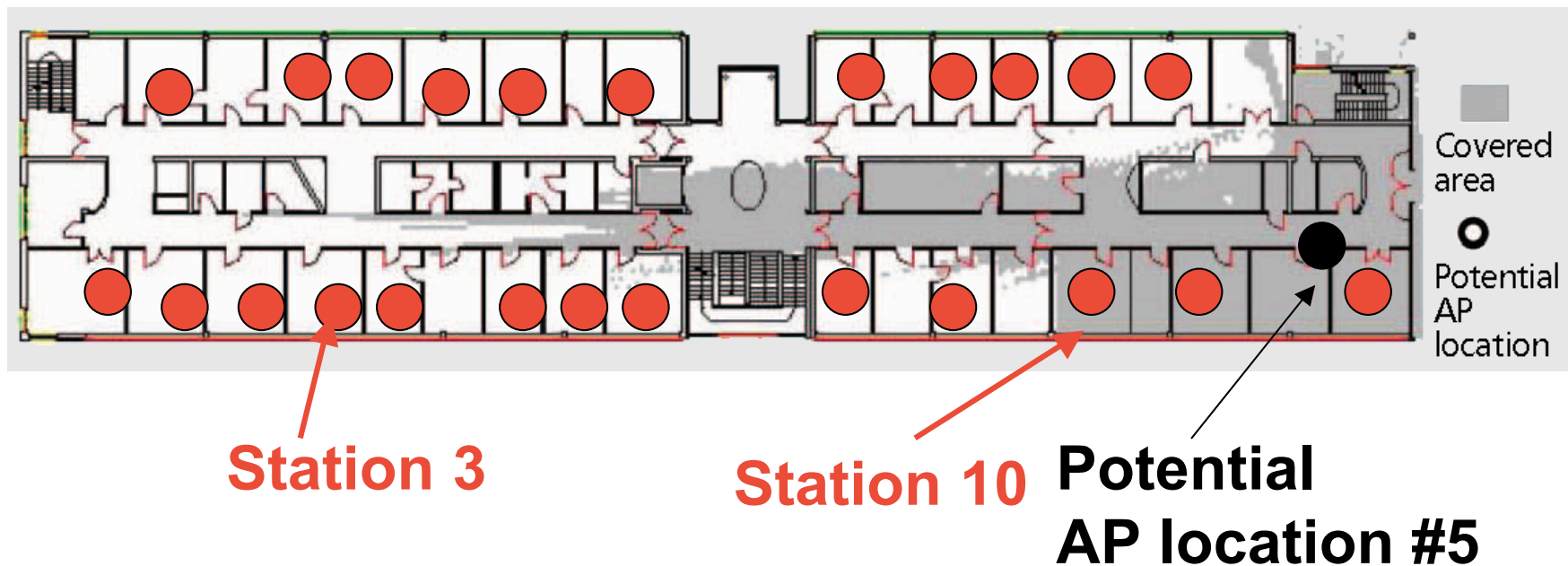
See the next page for a graphical explanation for δ_{ij}

- Find the minimum number of access points required so that
 - All stations are covered

Explanation of δ_{ij}

- Example: $\delta_{10,5} = 1$, $\delta_{3,5} = 0$

● = location of station



The coverage problem: Verbal formulation

Decision variables:

$$x_j = \begin{cases} 1 & \text{if an AP is to be installed at location } L_j \\ 0 & \text{otherwise} \end{cases}$$

Integer programming formulation:

min The number of access points (An expression in x_j)

subject to

Each station is covered (One expression for each station, need x_j and $\delta_{i,j}$)

$$x_j \in \{0, 1\}$$

The coverage problem: Formulation

Decision variables:

$$x_j = \begin{cases} 1 & \text{if an AP is to be installed at location } L_j \\ 0 & \text{otherwise} \end{cases}$$

Integer programming formulation:

$$\min \sum_{j=1}^p x_j$$

subject to

$$\sum_{j=1}^p \delta_{ij} x_j \geq 1 \quad \forall i = 1, \dots, n$$

$$x_j \in \{0, 1\}$$

Some alternative constraints

- Each station can associate with at least b AP

$$\sum_{j=1}^p \delta_{ij} x_j \geq b \quad \forall i = 1, \dots, n$$

- Each station can associate with at least c APs and at most d APs where $d \geq c$

$$\sum_{j=1}^p \delta_{ij} x_j \geq c \quad \forall i = 1, \dots, n$$

$$\sum_{j=1}^p \delta_{ij} x_j \leq d \quad \forall i = 1, \dots, n$$

Summary

- Placement problem
 - Using binary decision variables to decide whether to use a potential location or not
 - Important class of problems for (mixed) integer programming

Acknowledgments

- Picture credits:

- Andreas Eisenblätter and Hans-Florian Geerdes, "Wireless Network Design: Solution-oriented modeling and mathematical optimization", *IEEE Wireless Communications*, December 2006